CMOS Inverter – Week 3

Fire EV Internship | CMOS & VLSI

During this week, I learned about the working of a CMOS inverter using PMOS and NMOS transistors. I studied its static and dynamic behaviour, voltage transfer characteristic (VTC), switching threshold and noise margins. I also learned about propagation delay, rise time, fall time and power dissipation.

For practical work, I used Xschem to work on the CMOS inverter schematic and ngspice to study the circuit response. I generated the VTC and transient response plots and reviewed how transistor sizing affects inverter behaviour. This week helped me understand how a CMOS inverter switches and how its performance can be analysed.

Checkpoint Questions and Answers

Q1. What happens when the input of a CMOS inverter is LOW?
The PMOS turns ON and the NMOS turns OFF, so the output becomes HIGH.

Q2. What is a voltage transfer characteristic (VTC)?
It is a graph of output voltage (VOUT) against input voltage (VIN). It shows how the inverter output changes as the input voltage increases.

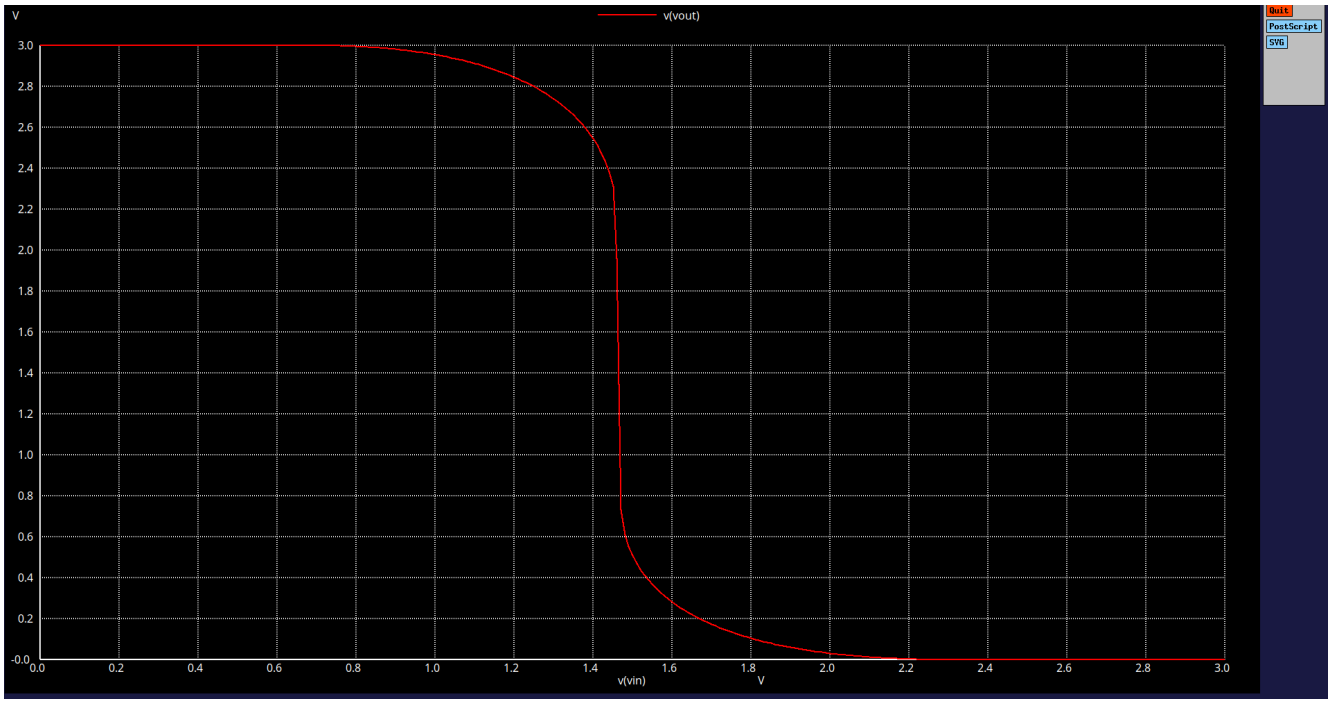
Q3. What are noise margins?
Noise margins indicate how much unwanted voltage disturbance a digital signal can tolerate without being interpreted incorrectly.

Q4. What is propagation delay?
It is the time taken for the output to respond to a change in the input.

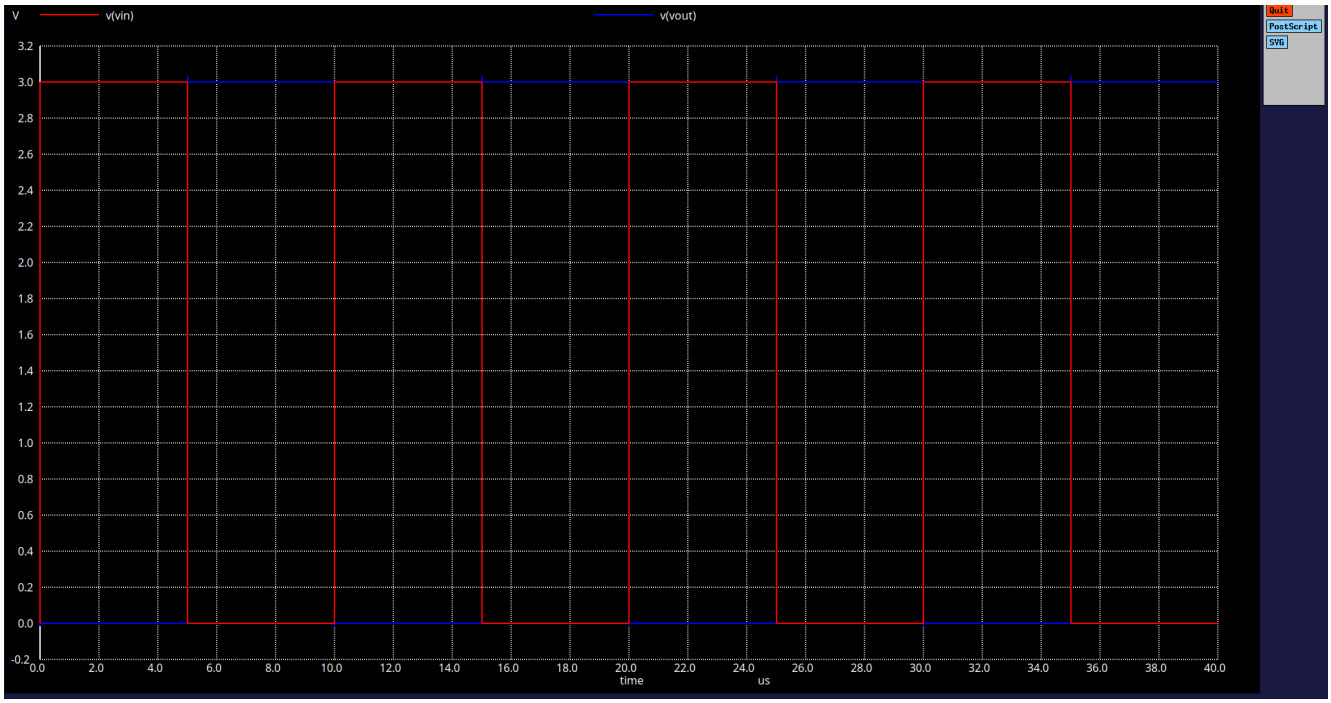
Q5. What are the main types of power dissipation in CMOS?
Static power, dynamic switching power and short-circuit power.

Simulation Plots

The following graphs were captured during the CMOS inverter simulation work.



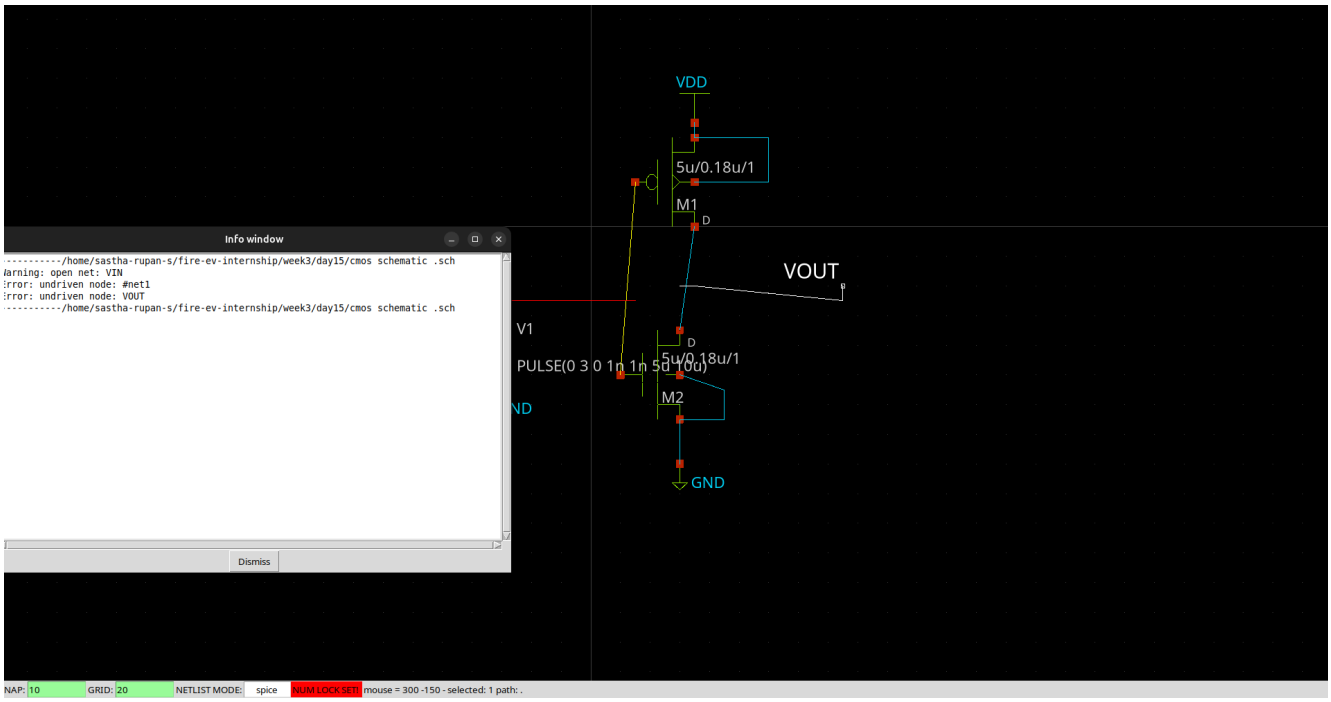
CMOS Inverter Voltage Transfer Characteristic – VOUT versus VIN.



CMOS Inverter Transient Response – input and output voltage waveforms over time.

Xschem Schematic

The screenshot below shows the CMOS inverter schematic work in Xschem, including the PMOS and NMOS devices, supply, ground, input source and output connection.



CMOS inverter schematic captured in Xschem.

Weekly Summary

This week, I learned the operation of a CMOS inverter, studied its VTC and noise margins, and explored its transient response and power dissipation. I also gained practice with Xschem and ngspice for circuit design and simulation.